A Thin Electroadhesive Inchworm Climbing Robot Driven by an Electrostatic Film Actuator for Inspection in a Narrow Gap

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Abstract—To inspect in a narrow gap of a critical construction or a large machine, such as a generator in a power plant, a thin inchworm climbing robot is proposed. It achieves the unique feature of low height by employing electrostatic adhesion and an electrostatic film actuator, both of which are composed of flexible printed circuit films. A prototype was developed in this research, which has only 2.4 mm of thickness and weighs 21.7 g. The experiments demonstrated that it can climb up a slope of 80°, and carry a 0.5 N payload on a horizontal surface with applied voltage of 600 V.

Keywords—Electroadhesion; electrostatic actuator; inchworm robot; climbing robot; narrow air gap

I. INTRODUCTION

In the field of safety and security robotics, climbing robot is a promising technique owing to its unique mobile capability. It can serve different tasks, especially inspection of high constructions or critical large machines. For such purposes, a number of climbing robots have been reported in literatures, which targeted, for example, inspection of nuclear power plants [1] or cable inspections of the cable-stayed bridge [2]. Other examples of applications include inspection of aircrafts [3], ships [4] and oil tanks [5].

Besides these applications, there are some tasks with more special requirements. In a particular situation, the height of the space for the mobility of robots is really limited. For example, as examining the stator of a generator or a gas turbine in a power plant, the air gap for free movement is considerably small (lower than 10 mm) [6][7]. Such applications require a climbing robot with considerably low height profile. There are, however, two main technical difficulties in the development of low height climbing robots, actuation method for locomotion and adhesion method. This paper focuses on the two promising technologies, electrostatic actuation and electrostatic adhesion, to fulfill the demand.

In previous studies, a climbing robot always relied on electromagnetic motors. As electromagnetic motors consist of ferromagnetic yokes, coils and permanent magnets, which require certain volumes to function magnetically, it is not easy to reduce their thicknesses. Resultantly, climbing robots with electromagnetic motors are limited in their minimum heights, due to the limited thinness of the employed electromagnetic motors. On the other hand, electrostatic actuators are comprised of conductive electrodes that can function with considerably small thickness [8]. Therefore, electrostatic actuators can be configured in a very thin structure. Hence, instead of using electromagnetic motors, this work introduces a thin electrostatic film actuator into the climbing robot for operation in a narrow space. Regarding adhesion methods [9][10][11], electrostatic adhesion becomes a suitable choice to fulfill the application requirement, as it is also suitable for thin structure and is compatible with the above actuation choice; electroadhesion also employs thin electrodes as same as the electrostatic actuation, which ensures the low-height profile of the robot [12][13].

Considering the structural similarity between electrostatic actuation and electrostatic adhesion, the integration of these technologies for climbing robots seems a natural choice, but such integration has not been studied extensively. Our previous paper [14] reported the first integration of the two electrostatic technologies, by employing a crawler type configuration. One of the prototypes reported in the paper could carry considerably heavy load (around 3 N) compared to its light weight (50 g). Although the reported crawler robot was compact (14 mm in height) compared to the most of other climbing robots, it still cannot access into the above-mentioned narrow space. It is, of course, true that the crawler robot can be designed lower in height than the reported prototype, but the minimum height exists to avoid plastic deformation of the electrode belt, which is imperative for the crawler robot. Therefore, to realize thinner structure, a different configuration needs to be investigated.

Although the previous report studies crawler type, climbing robots that integrate electrostatic actuation and electrostatic adhesion can be configured in many other different ways. This work focuses on another configuration, which is inchworm type, as it seems more suitable for the movement in a space with height limitation. With a careful design, an inchworm climbing robot can achieve considerably low height, which is even close to the thickness of an electrode film. For example, in this work, we realized a proof-of-concept prototype with

978-1-4799-0880-6/13/$31.00 ©2013 IEEE

height lower than 3 mm (including the diameter of feeding wires). Moreover, whereas the crawler prototypes reported in our previous work had limited motion strokes due to the connection of power feeding wires to the rotating electrode belt, the inchworm type does not suffer from such a technical problem owing to the reciprocating linear locomotion of the actuators. In addition, it can be lighter than the crawler robot, since an inchworm robot does not need a supporting structure which is necessary in the crawler robot. Such feature would also lead to mechanical flexibility of the robot to comply with curved surfaces.

In this paper, we propose and investigate an inchworm type climbing robot that integrates the electrostatic actuation and the electrostatic adhesion. In the next section, the basic principles of electrostatic actuation and electroadhesion will be briefly reviewed. Based on the principles, an inchworm climbing robot that is fully actuated by electrostatic force is proposed in Section III. Then Section IV reports on a simple proof-of-concept prototype that was designed and manufactured in this work. Finally, Section V describes the experiments that were conducted to test the climbing capacity, payload capacity and the velocity of the prototype

II. ELECTROSTATIC ACTUATION AND ELECTROADHESION

A. Electrostatic Actuation

Electrostatic actuators are well known in the field of micro-scale robotics. In this research, however, an electrostatic actuator that can work in an ordinary macroscopic scale is introduced to the climbing robot. Various types of macroscopic electrostatic actuators have been so far proposed in literatures, and here we employ an actuator that is called dual excitation multiphase electrostatic drive [8]. The actuator is made of a set of a stator electrode film and a slider electrode film, both of which are made using flexible printed circuit (FPC) technologies. Two pieces of the electrode films shown in Fig. 1(a) are used in this work for electrostatic actuator. The electrode films contain three-phase parallel straight electrodes. The electrodes, 0.08 mm in width, are aligned with a regular pitch of 0.16 mm. The surfaces of the electrodes are covered with insulation layer. For actuator operation, glass beads of 0.02 mm in diameter are scattered between the slider and the stator to reduce the friction force and to avoid stiction. Although in previous works on this electrostatic actuator [8], including our earlier crawler prototypes [14], insulating liquid was injected into the gap between the stator and the slider to enhance the output performance, this work does not utilize the insulating liquid.

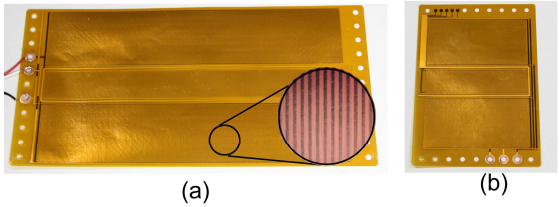


Fig. 1 The FPC films with three-phase electrode used in this work. The film in (a) was used for actuation (two identical films were used as the stator and the slider for Fig. 3, whereas one film was cut into shorter length for the prototype). The film in (b) was used for electroadhesion.

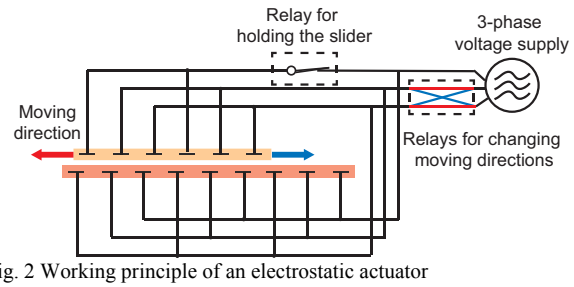


Fig. 2 Working principle of an electrostatic actuator

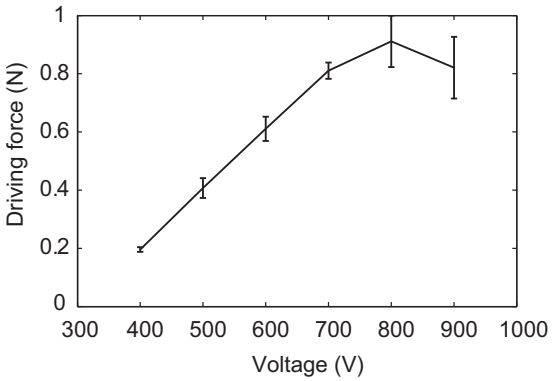


Fig. 3 The driving force of the electrostatic actuator obtained in the atmospheric condition using the electrode films shown in Fig. 1(a).

While supplying a three-phase high voltage to both the slider and the stator with reversed phase orders to each other, electrostatic thrust force will be generated to drive the slider film, as shown in Fig. 2. The operation of the actuator is synchronous; the moving speed can be calculated by the voltage frequency as

$$u = 6pf \tag{1}$$

where u is the speed, p is the electrode alignment pitch (0.16 mm in this work), and f is the frequency. If the three-phase high voltage changes to the different order by relays, the moving direction of the slider will be opposite, as shown in Fig. 2. When one of the three phases is failed to be connected to the stator or the slider, the actuator cannot generate proper thrust force and will stop with the two components sticking to each other. We will employ these phenomena to control the locomotion of the inchworm robot.

Its thrust force was tested with two pieces of the electrode film shown in the Fig. 1(a) in the atmospheric condition. The results are shown in Fig. 3. In the previous reports on the electrostatic actuator, the actuator was typically driven within dielectric liquid that allows use of high voltage. However, the use of a dielectric liquid is not suitable for the climbing robots and thus the operation has been done in the atmosphere in this work, which limited the applied voltage due to corona discharge within the air gap between the stator and the slider. The plot in Fig. 3 shows that thrust force increased with the increase of the applied voltage but peaked out around the applied voltage of 800 V, which would probably be a result of the corona discharge.

B. *Electroadhesion*

Electrostatic adhesion is realized by the electrostatic attraction force between induced charges on a substrate and surface-insulated electrodes. As shown in Fig. 4(a), as the voltage is supplied to the electrodes, electric charges are induced on the surface of the substrate, which is typically, but not necessarily limited to, conductive. The attraction force between the electrode and the substrate can be calculated using a simple text-book equation as

$$F_{ad} = \frac{1}{2}\epsilon S\left(\frac{V}{d}\right)^2 \quad (2)$$

where ε is the permittivity, S is the area of electrodes, V is the applied voltage, and d is the gap between electrodes and the substrate surface.

For electroadhesion in this paper, we employ the electrode film with three-phase electrodes shown in Fig. 1(b). Its fundamental structure is the same as the FPC film used for the electrostatic actuator. (The film in Fig. 1(b) has skewed electrodes [15], but this should not have any effect on adhesion.) The use of three-phase electrodes allows us to share the same three-phase high voltage for the actuator and hence reduces the size and cost of the driving circuit.

The attraction force generated by application of three-phase sinus voltage to the three-phase electrodes, shown in Fig. 4(b), is calculated as [14]

$$F_{ad-3} = \frac{1}{2}\epsilon S\left(\frac{V}{d}\right)^2\left(\sin^2\theta + \sin^2\left(\theta - \frac{2}{3}\pi\right) + \sin^2\left(\theta + \frac{2}{3}\pi\right)\right)$$
$$= \frac{3}{4}\epsilon S\left(\frac{V}{d}\right)^2 \quad (3)$$

where θ is the phase of the applied three phases. While the electrode is adhered to the substrate, the friction force is caused by the attractive force as

$$F_{f-ad} = \mu F_{ad-3} \quad (4)$$

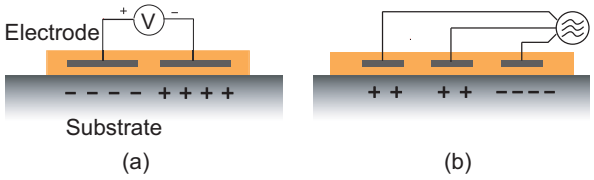


Fig. 4 The basic principle of electroadhesion with two-phase voltage (a) and with three-phase voltage (b)

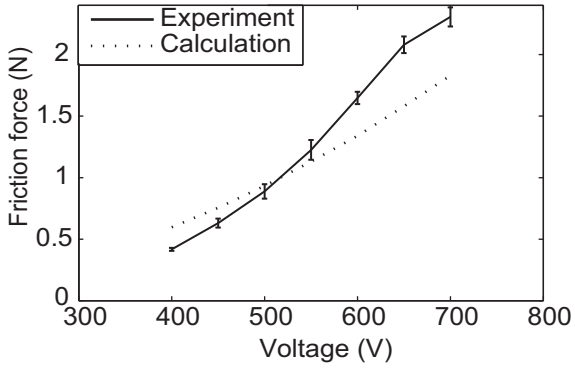


Fig. 5 The friction force caused by the electrostatic adhesion force obtained using the electrode film shown in Fig. 1 (b).

where μ is the coefficient of the friction force. Fig. 5 shows the measured friction force with different applied voltages using the electrode film shown in Fig. 1(b). In this experiment, the electrode film supplied with three-phase voltage was situated on an aluminum board and pulled by a force sensor in the parallel direction to the surface of the board. The plot in Fig. 5 also shows the theoretical results calculated using Eq. (4). They are close but have a little difference. When the voltage is lower than 500 V, the experimental data are lower than the calculation results, probably because of the increased air gap between the electrode film and the substrate surface due to the intrinsic deformation of the film. When the voltage is higher than 500 V, on the other hand, the experimental results are higher than the calculation. It is probably because the adhesion force was strong enough to produce the air pressure (suction cup) effect [12].

III. THE CONCEPT OF THE THIN INCHWORM CLIMBING ROBOT BASED ON ELECTROADHESION AND ELECTROSTATIC ACTUATION

Based on the principles of electrostatic actuation and electrostatic adhesion, a climbing robot with inchworm locomotion can be built. One of the possible structures is shown in Fig. 6. The inchworm robot is composed of a main supporting body and a series of moving components which we call feet. All of the components are made of electrode films. A foot component consists of two electrode films; one film is employed as a slider of an electrostatic actuator while the other is utilized for adhesion.

Owing to the use of electrode films as components in both actuation and adhesion, an inchworm robot can be significantly thin. For example, we realized a prototype with height lower than 3 mm (including the diameter of power feeding wires), as introduced in the next section. Because of the same reason, the robot can be considerably flexible. All of its components, including the electrode films and the feeding wires, are expected to be flexible, which should allow the whole robot body to deform to comply with curved surfaces as shown in Fig. 7.

The inchworm robot with multiple feet can be controlled with different strategies for different tasks. By coordinating the operation periods of actuation and adhesion, each foot component of the robot can move forward or serve as a supporting foot to drive the main body of the robot. For

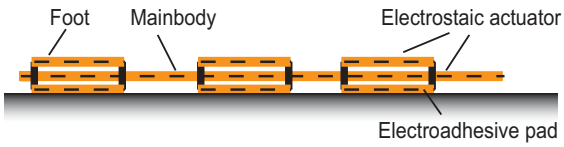


Fig.6 The structure of an inchworm climbing robot

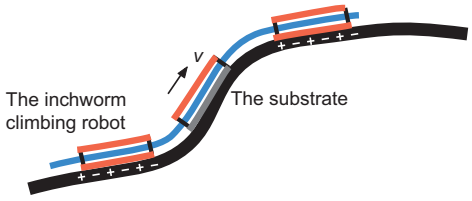


Fig. 7 An inchworm robot is climbing on a curved surface

example, assuming the payload is connected to the main body, the robot can drive the first, second, and the third foot one by one, and then finally drive the main body so that it can carry heavy payload (but with low velocity), as shown in Fig. 8 (a). In another case where the payload is not so heavy, it can drive the first and the third foot together, which is followed by the actuation of the second foot and the main body, as shown in Fig.8 (b). This driving strategy is expected to realize faster motion but with lower payload capacity.

IV. Mechanical Structure Design and Locomotion Control of the Robot

A. Mechanical structure design

To verify the proposal of the electrostatic inchworm climbing robot, a simple prototype with two feet was developed. The two feet are referred to as the forefoot and the hindfoot, as shown in Fig. 9. In this simple prototype, both of the forefoot and the hindfoot have adhesion function, but the actuation function is given to only the forefoot; the hindfoot is fixed to the main body. The forefoot has two electrode films which sandwiches the main body. The upper electrode film functions as a slider of electrostatic actuator allowing the foot to move along the main body, and the lower film functions as adhesive electrode. A sheet of polyimide film was folded to wrap around the two electrode films of the actuator and connected to the slider. The folded edges of the wrapping polyimide film also functions as a linear guide for the electrostatic actuator.

The main body that functions as the stator of the actuator is the FPC film shown in Fig. 1 (a), whereas the slider film in the forefoot was cut out from the FPC film shown in Fig. 1 (a) to have shorter length. The length of the effective electrode areas measure 171 mm for the stator, and 128 mm for the slider. The hindfoot, which is glued to the main body, has only one electrode film shown in Fig. 1 (b), which is used for adhesion. The appearance of the robot is shown in Fig. 10. Its height is only 2.4 mm resulting from its film structure. Its width is 120 mm, and the length is 235 mm. The total weight is 21.7 g.

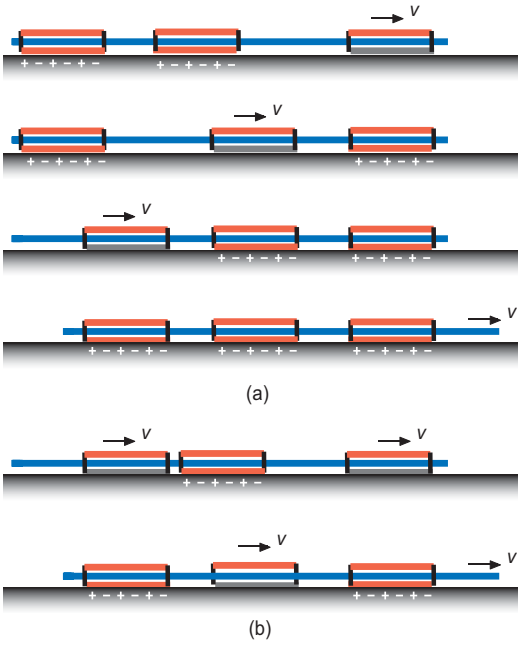


Fig.8 The high-payload locomotion pattern (a) and the fast locomotion pattern (b) of the inchworm climbing robot with many joints

B. Locomotion control

As the proof-of-concept prototype is realized with only two feet, one of which is fixed to the main body, it cannot be controlled by the proposed strategies shown in Fig. 8 which assumes multiple feet. Thus, another simpler locomotion strategy was adopted for this prototype, which is shown in Fig. 11.

First, in the "stop" phase, the adhesive films in both of the forefoot and the hindfoot are activated with high voltages so that the whole robot adheres to the substrate. In this phase, the actuator has one disconnected phase that makes the slider stick to the main body.

In the next phase, "moving phase 1", the adhesive electrode in the forefoot is deactivated by cutting the connection to the high-voltage supply, which allows the forefoot to move freely against the main body. At the same time, all the three phases of the actuator are connected to the high-voltage supply, which actuates the forefoot to move forward against the main body that remains stationary.

After the first moving phase, the "stop" phase is repeated so that the robot stops again to make the forefoot to adhere to the

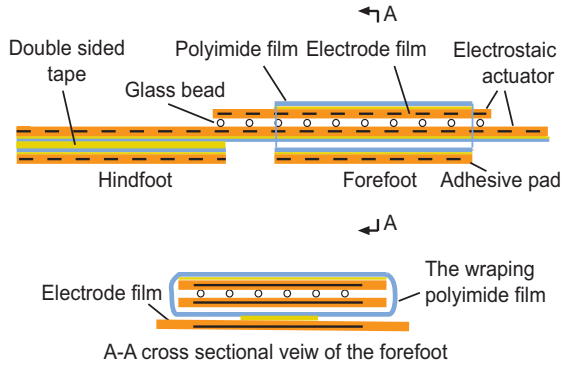


Fig. 9 The structure of the prototype inchworm climbing robot

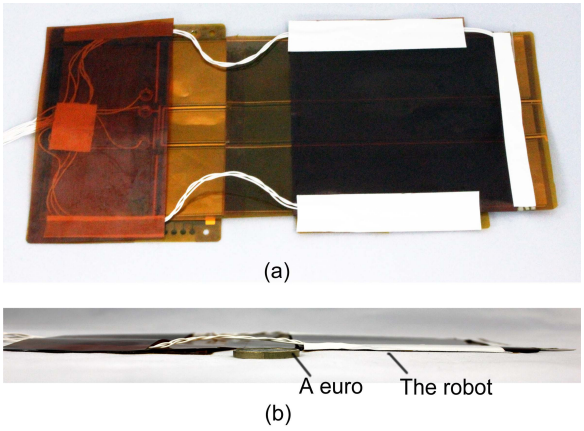


Fig. 10 The inchworm climbing robot (a) and its side view (b)

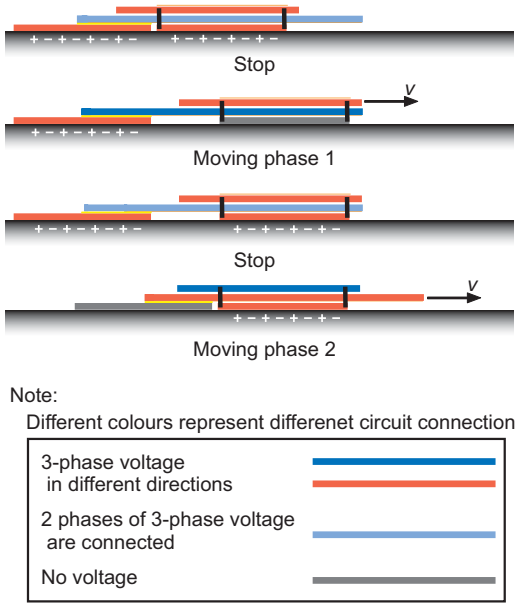


Fig. 11 A locomotion cycle of the inchworm climbing robot

substrate, together with the hindfoot.

Then in the second moving phase, "moving phase 2", the adhesive electrode in the forefoot is kept activated, while the adhesive electrode in the hindfoot is deactivated. Meanwhile, the electrostatic actuator in the forefoot is driven to move in the opposite direction to "moving phase 1". With this operation, the forefoot remains at the stationary place, whereas the whole body moves forward by the electrostatic actuation.

This is just one cycle of the locomotion of the developed prototype inchworm robot. By repeating this cycle again and again, the inchworm robot can move along a substrate with adhering on it by electrostatic adhesion.

To realize the above locomotion, this work used one set of three-phase voltage suppliers, which continuously generates a three-phase high voltage. For controlling the motion of the robot, a PIC controller (PIC18F4550) together with photo-MOS relays (AQV258) switched the connections between the high-voltage supplies and the electrodes of the robot. It controlled the connecting order to change the moving direction of the actuator, as well as the connection time for the proper movement periods.

The moving direction of electrostatic actuator is decided by the connection of the three-phase voltage to the three-phase electrodes of the stator and the slider. The moving direction is reversed by changing the connecting order of the three phases, even if the three-phase voltage remains the same. The PIC controller was programmed to switch the connecting order depending on the motions of the forefoot. To hold the forefoot against the stator, the PIC controller cuts the connection of one phase of the three-phase high voltage. By cutting the connection of one phase, the actuator loses actuation function and the slider simply sticks to the stator.

The PIC controller also controls the voltage application to the adhesive films by connecting or disconnecting the three-phase voltage to the three-phase electrode of the adhesive films. The voltage in the control circuit including the PIC controller and the control side of the photo-MOS relay was 5 V, whereas the driving high voltage connected to the robot for electrostatic actuation and electrostatic adhesion was up to 600 V in its amplitude. The frequency of the high-voltage three-phase was fixed to 5 Hz.

V. Experimental Result

A. Climbing on a slope

In Section II, the data for the friction force caused by the electroadhesion showed that the robot with weight of 21.7 g could be supported even on a vertical slope with the voltage of over 400 V. To verify the performance of electroadhesion of the inchworm robot, the prototype was tested, at first, on a slope of 80°. The setup is shown in Fig. 12. An aluminum board was utilized as a slope to be climbed up. The inchworm robot was driven with voltage of 500 V and 5 Hz. The experimental results confirmed that the prototype can climb up a slope of 80°. However, when the prototype was tested on a 90° slope, the robot failed to operate because the thin robot deformed away from the substrate during the forefoot was moving up.

B. Payload and velocity

The payload capacity of the robot was tested on a horizontal surface of an aluminum board with a force sensor connected to the forefoot. The robot was driven with the voltage of lower than 600 V and the frequency of 5 Hz.

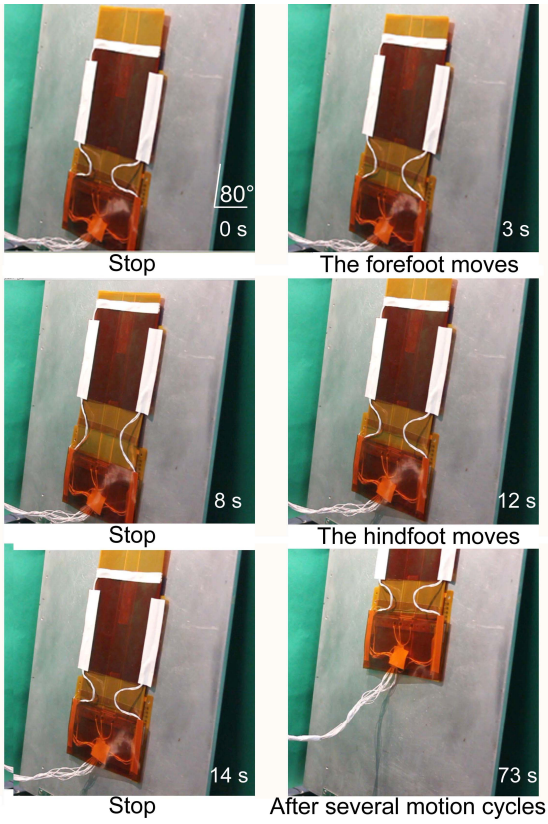


Fig. 12 The process when the inchworm climbing robot is climbing up an 80° slope

The payload capacity of the robot is decided by its driving force and the adhesion force, but in this prototype it is limited by the former one as it is weaker than the adhesion force as described in Section II. The result shown in Fig. 13 reveals that the robot can carry a 0.5 N payload in maximum at the voltage amplitude of 600 V. Fig. 13 also shows the estimated performance, which was calculated from the results of Fig. 3 by multiplying 0.75, which is the ratio of the effective electrode areas between the prototype robot and the actuator used for Fig. 3. The comparison indicates that the experiment data are close to the estimated values.

The velocity was measured using a laser displacement sensor, on a horizontal surface without external load. To facilitate the measurement, a reflecting marker was installed on the hindfoot. The result is shown in Fig. 14. Although the theoretical synchronous velocity of the electrostatic actuator is 4.8 mm/s at the frequency of 5 Hz, the measured actuator velocity was around 4.7 mm/s. This means that a little slip existed between the robot and the surface even without load. Regarding the total average velocity of the robot, it was found around 1.5 mm/s, which is much slower than the actuator's velocity since the forefoot and the hindfoot of the robot moves alternatively with stop phases. The total velocity can be improved by reducing the duration of the stop phase and/or increasing the frequency of three-phase voltage.

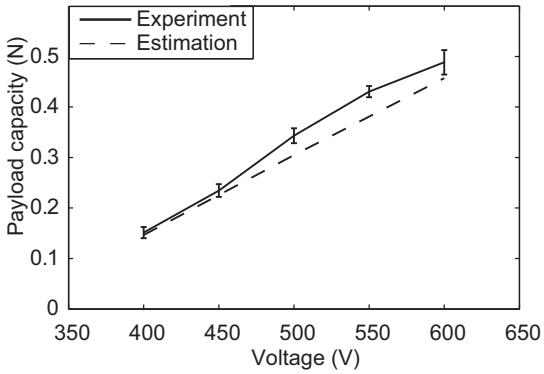


Fig. 13 The payload capacity of the inchworm climbing robot and the estimation from the experiment data of driving force in Section II (scaled to the same area size)

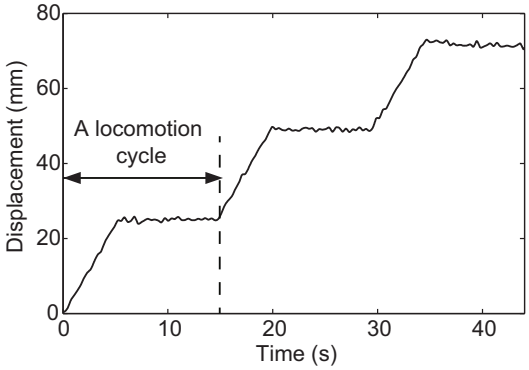


Fig. 14 The displacement of the hindfoot of the inchworm climbing robot

VI. Conclusions

To inspect a narrow gap, this paper proposed a new kind of low-height inchworm electroadhesive climbing robots. It is distinctive because of its appearance of its film-like shape, which is considerably different from previous climbing robots. The mechanism of the climbing robot also makes it light in weight. These characteristics are promising for inspection in a narrow gap.

The proof-of-concept prototype was designed with minimum components and fabricated in this work. It measures only 2.4 mm in its height, and weighs 21.7 g. The experiments showed that as a climbing robot, it can climb up a slope up to 80° with electrostatic adhesion and carry a 0.5 N payload with the voltage amplitude of 600 V.

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